培道中學校

姓名：廖汝鋒　　（　號）

年級：微91

科目：概率論與隨機過程

成績

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1. Solution：$\int_{-\infty}^{+\infty} f_X(x)\,dx = \int_{-\infty}^{+\infty} \frac{\lambda}{2} e^{-\lambda|x|}\,dx = 2\times\frac{\lambda}{2}\int_0^{+\infty} e^{-\lambda|x|}\,dx = \int_0^{+\infty} e^{-\lambda x}\,d(\lambda x)$

$= e^{-\lambda x}\Big|_{+\infty}^{0} = 1 \Rightarrow f_X$ satisfies the normalization condition

$$E[X] = \int_{-\infty}^{+\infty} x\cdot\frac{\lambda}{2} e^{-\lambda|x|}\,dx = 0$$

$$\mathrm{var}(X) = E[X^2] - (E[X])^2 = E[X^2] = \int_{-\infty}^{+\infty} x^2 \frac{\lambda}{2} e^{-\lambda|x|}\,dx$$

$$= \int_0^{+\infty} x^2 e^{-\lambda x}\,d(\lambda x) = x^2 e^{-\lambda x}\Big|_{+\infty}^{0} - \int_{+\infty}^{0} e^{-\lambda x}\cdot 2x\,dx$$

$$= \int_0^{+\infty} 2x e^{-\lambda x}\,dx = \frac{2}{\lambda}\left[x e^{-\lambda x}\Big|_{+\infty}^{0} - \int_{+\infty}^{0} e^{-\lambda x}\,dx\right] = \frac{2}{\lambda}\int_0^{+\infty} e^{-\lambda x}\,dx$$

$$= \frac{2}{\lambda^2}$$

2. (a) Obviously, $\frac{f_X(x)}{2\pi x}$ is a constant C. (When $0\le x\le r$)

Also, $C\cdot\pi r^2 = 1 \Leftrightarrow \frac{1}{\pi r^2} = C = \frac{f_X(x)}{2\pi x} \Leftrightarrow f_X(x) = \frac{2}{r^2}x$ (When $0\le x\le r$)

$$\Rightarrow f_X(x) = \begin{cases} \frac{2}{r^2}x, & 0\le x\le r \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \int_0^r x\cdot\frac{2}{r^2}x\,dx = \frac{2}{3r^2}r^3 = \frac{2}{3}r$$

$$\mathrm{var}(X) = E[X^2] - (E[X])^2 = \int_0^r x^2\cdot\frac{2}{r^2}x\,dx - \frac{4}{9}r^2 = \frac{1}{2}r^2 - \frac{4}{9}r^2 = \frac{1}{18}r^2$$

~~(b) When $s<0$, $f_S(s)=0 \Rightarrow F_S(s)=0$~~

~~When $s\ge 0$, $f_S(s)=$~~

(b) When $s<0$, $F_S(s)=0$

when $0\le s\le \frac{1}{t}$, $F_S(s) = P(r\ge X\ge t) = \int_t^r \frac{2}{r^2}x\,dx = 1-\frac{t^2}{r^2}$

when $s>\frac{1}{t}$, $F_S(s) = P(X\ge \frac{1}{s}) = \int_{\frac{1}{s}}^{r}\frac{2}{r^2}x\,dx = 1-\frac{1}{(sr)^2}$

$$F_S(s) = \begin{cases} 0, & s<0 \\ 1-\frac{t^2}{r^2}, & 0\le s\le\frac{1}{t} \\ 1-\frac{1}{(sr)^2}, & s>\frac{1}{t}\end{cases}$$

(S isn't a C.R.V).

3. P(a reading of 5 $\cap$ the point belongs black area) = P(a reading of 5 $\cap$ white area)

$$\Leftrightarrow \alpha \cdot \frac{1}{3\sqrt{2\pi}} \cdot e^{-\frac{(5-6)^2}{2\times 9}} = (1-\alpha) \cdot \frac{1}{2\sqrt{2\pi}} \cdot e^{-\frac{(5-4)^2}{2\times 4}}$$

$$\Leftrightarrow \frac{\alpha}{3\times e^{\frac{1}{18}}} = \frac{1-\alpha}{2\times e^{\frac{1}{8}}} \Leftrightarrow \alpha = \frac{3\times e^{\frac{1}{18}}}{3\times e^{\frac{1}{18}} + 2\times e^{\frac{1}{8}}}$$

4. $$\Gamma\left(\tfrac{1}{2}\right) = \int_0^{\infty} e^{-x} x^{\frac{1}{2}-1}\, dx = \int_0^{\infty} \frac{e^{-x}}{\sqrt{x}}\, dx = \int_0^{\infty} \frac{e^{-t^2}}{t}\, d(t^2) = 2\int_0^{\infty} e^{-t^2}\, dt$$

$$= \int_{-\infty}^{+\infty} e^{-x^2}\, dx = \int_{-\infty}^{+\infty} e^{-\left(\frac{x}{\sqrt{2}}\right)^2} d\left(\frac{x}{\sqrt{2}}\right) = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2}} e^{-\frac{x^2}{2}}\, dx = \sqrt{\pi}\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dx$$

$N(0,1)$: $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

$$\Rightarrow \Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}\int_{-\infty}^{+\infty} f_X(x)\, dx = \sqrt{\pi}$$